Symbols used in 'dirac16complex in the primordial, modern and future gravitational fields'

symbol	meaning	definition
$x_0, \ldots, x_7$	coordinates	x_0 hidden, $x_{1,2,3}$ space, x_4 time, $x_{5,6,7}$ superluminal time
η_{AB}	flat metric	$\mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$
T^A	real 16x16 Dirac matrices $(T16^A)[A]$	$T^A T^B + T^B T^A = 2\eta^{AB} I_{16}$
T_A	lowered index	$T_A = \eta_{AB} T^B$
Γ	chirality	$T^0 \cdots T^7 = \mathrm{diag}(-I_8, I_8)$
σ_{16}	Dirac-adjoint matrix	$T^0 T^1 T^2 T^3$
ψ, $\bar{\psi}$	dirac16complex and adjoint	complex Grassmann 16-spinor; $\bar{\psi} = \psi^\dagger \sigma_{16}$
${e^A}_\mu$, ${E_A}^\mu$	vielbein and inverse	$g_{\mu\nu} = \eta_{AB} {e^A}_\mu {e^B}_\nu$, ${E_A}^\mu {e^A}_\nu = \delta^\mu_\nu$
e_A	diagonal vielbein entries	$e = \mathrm{diag}(e_0, \ldots, e_7)$
$g_{\mu\nu}$, $\lvert g\rvert$	metric, absolute determinant	$\sqrt{\lvert g\rvert} = \prod_A \lvert e_A\rvert$
${\Gamma^\nu}_{\mu\lambda}$	Christoffel symbols	$\frac{1}{2} g^{\nu\kappa}(\partial_\mu g_{\kappa\lambda} + \partial_\lambda g_{\kappa\mu} - \partial_\kappa g_{\mu\lambda})$
${\omega_\mu}^A{}_B$	spin connection	${e^A}_\nu(\partial_\mu {E_B}^\nu + {\Gamma^\nu}_{\mu\lambda} {E_B}^\lambda)$
Ω_μ	spinor connection	$\frac{1}{4} {\omega_\mu}^A{}_B T_A T^B$
γ^μ	curved Dirac matrices	${E_A}^\mu T^A$
D_μ	covariant derivatives	$D_\mu \psi = \partial_\mu \psi + \Omega_\mu \psi$, $D_\mu \bar{\psi} = \partial_\mu \bar{\psi} - \bar{\psi}\Omega_\mu$
$\mathcal{L}$	Lagrangian density	$\sqrt{\lvert g\rvert}[\frac{1}{2}(\bar{\psi}\gamma^\mu D_\mu \psi - (D_\mu \bar{\psi})\gamma^\mu \psi) - m\bar{\psi}\psi - \frac{\lambda}{2}(\bar{\psi}\psi)^2]$
m, λ	mass, quartic coupling	real constants ($m = 1$, $\lambda = -0.03$ in the numerics)
H	Hubble parameter of the primordial field	$H = 1$ in the numerics
r_s	mass (Schwarzschild-type) parameter	$f = 1 - r_s/x_0$; $r_s = 1$ in the numerics
H_Λ	dark-energy Hubble rate	$f = 1 - r_s/x_0 - H_\Lambda^2 x_0^2$; $H_\Lambda = 0.1$
f, f'	metric function and derivative	as above; $f' = df/dx_0$
e_P, e_M, e_F	primordial, modern, future vielbeins	Section 2
R_1	reflection	$x_1 \mapsto -x_1$
$\mathcal{C}$, $\mathcal{P}_1$	mass-sign maps	$\mathcal{C}\psi = \Gamma\psi^*$, $\mathcal{P}_1\psi(x) = T^1\psi(R_1 x)$
ρ_s	Kohn-Sham scalar density	$\langle\bar{\psi}\psi\rangle_{KS} = \sum_{occ} \bar{\phi}_n \phi_n$
ϕ_n	Kohn-Sham orbitals	eigenfunctions of h_{KS}
M	effective mass	$m + \lambda\rho_s$
B, N	lapse and hidden scale	$B = e_4$, $N = e_0$
ρ	tortoise coordinate	$d\rho = (N/B)dx_0$
ϵ	Kohn-Sham energy	$\psi = e^{-i\epsilon x_4} u(x_0)$
u, χ, (u, v)	rescaled orbitals	$u = (\sqrt{\lvert g\rvert}/N)^{-1/2}\chi$; $\chi = (u, v)$ in a 2×2 copy
h_{KS}	Kohn-Sham Hamiltonian	$\alpha p + W\beta$
α, β, p	Dirac operators	$\alpha = -T^4 T^0$, $\beta = -iT^4$, $p = -i\, d/d\rho$
W	effective potential	$W = BM$
A	2×2 copy operator	$A = d/d\rho + W$, $h = \begin{pmatrix} 0 & A^T \\ A & 0 \end{pmatrix}$
K	Krein operator	$-i\sigma_{16} T^4$; $\sigma_{16} = K\beta$
s_k	singular values of discretized A	Kohn-Sham levels $\pm s_k$
d	grid spacing	0.02
W_0	Lambert function	$W_0(y)e^{W_0(y)} = y$
ζ	vacuum label	$+1$ for V_1, -1 for V_2
V_1, V_2	Spin(4,4)-symmetric and zero-energy vacua	Section 4
ρ_{copy}, ρ_k	sea density of one copy, density of level k	$\rho_{copy} = -\sum_k \rho_k$
E_0, E_1	ground and first-excited total energies	ΔSCF
T	temperature	Fermi-Dirac occupations
$\Omega(T)$	thermodynamic effective potential	$8\sum_k(-s_k - 2T\ln(1 + e^{-s_k/T})) - \frac{\lambda}{2}\int B\rho_s^2 d\rho$
$\bar{M}$	mean effective mass	average of M over the grid
T_H	Hawking temperature	$1/(4\pi r_s)$ for the modern field
$\psi_\pm$, $S_\pm$, $Q_\pm$	mirror sectors, scalars, charges	$K\psi_\pm = \pm\psi_\pm$, $S_\pm = \psi_\pm^\dagger \beta \psi_\pm$, $Q_\pm = \int \psi_\pm^\dagger \psi_\pm$
$\mathcal{L}_D$	8-component Dirac Lagrangian	$\chi^\dagger \beta(T^A \partial_A - m)\chi$